

The Representation Technique for Exact Subset Sum: Reproducing the $2^{n/2}$ -to- $2^{0.291n}$ Frontier

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Abstract—The exact subset sum problem has a classical worst-case $\mathcal{O}^*(2^{n/2})$ meet-in-the-middle algorithm. The representation/dissection technique of Howgrave-Graham-Joux and Becker-Coron-Joux lowers the exponent to $\tilde{\mathcal{O}}(2^{0.291n})$ for *random* hard knapsacks, but not in the worst case. We reproduce the complexity models of both papers from their primary sources and numerically recover their published optima: the HGJ two-part ideal $2^{0.3113n}$, the HGJ simple algorithm $2^{0.337n}$ (May-Meurer corrected), and the BCJ three-level optimum $2^{0.291n}$. We then probe *where* the technique stalls: generic and dissociated instances have well-spread modular representations and no pseudo-solutions, while the open coefficient sets $\{0, 1\}$, $\{\pm 1\}$, $\{\pm 2\}$ are, within a bounded search ($C_1, C_2 \subseteq [-3, 3]$), exactly those that admit no nontrivial sumset factorisation, so the worst-case tool of coefficient shifting cannot be applied. No worst-case improvement is claimed.

Index Terms—subset sum, representation technique, meet-in-the-middle, dissection, hard knapsacks, reproducibility

I. INTRODUCTION

Given integers $a_1, \dots, a_n, b$, the exact subset sum problem asks whether some $x \in \{0, 1\}^n$ satisfies $\sum_i a_i x_i = b$. Horowitz-Sahni solve it in $\mathcal{O}^*(2^{n/2})$ time and space [1]; Schroeppe-Shamir keep the $\mathcal{O}^*(2^{n/2})$ time with $\mathcal{O}^*(2^{n/4})$ space [2]. Whether the exponent can be reduced below $1/2$ in the worst case is open. For *random* hard knapsacks of density $d = n/\log_2 \max_i a_i \approx 1$, the representation technique of Howgrave-Graham-Joux [3] and Becker-Coron-Joux [4] reaches $\tilde{\mathcal{O}}(2^{0.337n})$ and $\tilde{\mathcal{O}}(2^{0.291n})$ respectively; Chen et al. [5] give the first worst-case improvement, but only a polynomial factor $2^{n/2}/n^{0.5023}$. Most recently, Randolph-Wgrzycki [6] break the meet-in-the-middle barrier in the *worst case* for most constant coefficient sets, leaving $\{0, 1\}$, $\{\pm 1\}$ (Partition), and $\{\pm 2\}$ open.

This note does three things. It reproduces the HGJ and BCJ complexity models and their published optima from the primary sources (Sections III and IV); it probes which structural hypotheses about the barrier hold (Section V); and it characterises the open cases (Section VI). It makes no worst-case claim.

II. THE REPRESENTATION PRINCIPLE

Meet-in-the-middle splits the n entries into two halves L, R and looks for $u \in S(L), v \in S(R)$ with $u + v = b$ (split correctness; see the companion baselines note), at cost $\mathcal{O}^*(2^{n/2})$. The representation technique instead represents a solution $x \in \{0, 1\}^n$ in many ways as a sum of partial vectors,

so that a random modular filter keeps about one representation while shrinking the search.

For a solution of Hamming weight $\ell \approx n/2$, write $b = \sigma_1 + \sigma_2$ with σ_i a sum of $\lceil \ell/2 \rceil$ or $\lfloor \ell/2 \rfloor$ entries. The number of such decompositions is $N = \binom{\ell}{\lfloor \ell/2 \rfloor} \approx 2^\ell$. Choose a modulus $M \approx N$ and a random R ; heuristically some decomposition has $\sigma_1 \equiv R$ and $\sigma_2 \equiv b - R \pmod{M}$. Constructing the two subsets of solutions and matching them costs

$$D(\alpha) = \frac{2}{\alpha^{\alpha/2}(2-\alpha)^{(2-\alpha)/2}} \cdot 2^{-\alpha}, \quad (1)$$

where $\alpha = \ell/n$ and the cost is $D(\alpha)^n$; in the balanced case $D(1/2)^n = \tilde{\mathcal{O}}(2^{0.3113n})$. The remaining difficulty is to construct the modular solution lists in time proportional to their size; this is the crux.

A. Generalised coefficient sets

Becker-Coron-Joux extend the coefficient set from $\{0, 1\}$ to $\{-1, 0, 1\}$: a 1 is split as $(0, 1)$ or $(1, 0)$, and a 0 as $(0, 0)$, $(1, -1)$, or $(-1, 1)$. With a parameter α giving $(1/4 + \alpha)n$ ones and αn minus-ones per side, the number of representations is $N_D = \binom{n/2}{n/4} \cdot \binom{n/2}{\alpha n, \alpha n, (1/2 - 2\alpha)n}$.

III. REPRODUCING THE HGJ EXPONENTS

The HGJ simple algorithm [3] decomposes b into four parts $b = \sigma_1 + \sigma_2 + \sigma_3 + \sigma_4$, each a weight- $(n/8)$ sum, and solves each sub-knapsack with the *unbalanced* Schroeppe-Shamir algorithm. With modulus $M \approx 2^{\beta n}$, $1/4 < \beta < 1/3$, the paper's analysis gives

$$T(\beta) = \max(2^{0.3113n}, 2^{(0.0872 + \beta)n}), \quad M(\beta) = 2^{(0.5436 - \beta)n}. \quad (2)$$

The optimum is at $\beta \rightarrow 1/4$: time $2^{0.337n}$ with $2^{0.294n}$ memory. Other trade-offs are $\beta \rightarrow 1/3$ ($2^{0.421n}$ time, $2^{0.211n}$ memory) and $\beta \approx 0.2936$ (Schroeppe-Shamir memory $2^{n/4}$, time $2^{0.381n}$). The recursive variant (HGJ, Sec. 4.3) [3] applies the decomposition to the complementary knapsack near $\alpha = 1/2$; its thresholds are $\alpha_0 \approx 0.312835$ and $\alpha_1 \approx 0.458959$. Table I shows these values reproduced numerically.

TABLE I

REPRODUCED REPRESENTATION EXPONENTS (BASE-2, PER n), FROM THE PRIMARY-SOURCE MODELS [3], [4].

Quantity	Exponent
HGJ two-part ideal $D(1/2)$	0.3113
HGJ simple algorithm (best β)	0.337
BCJ $\alpha = \beta = \gamma = 0$ (HGJ)	0.337
BCJ minimised	0.291

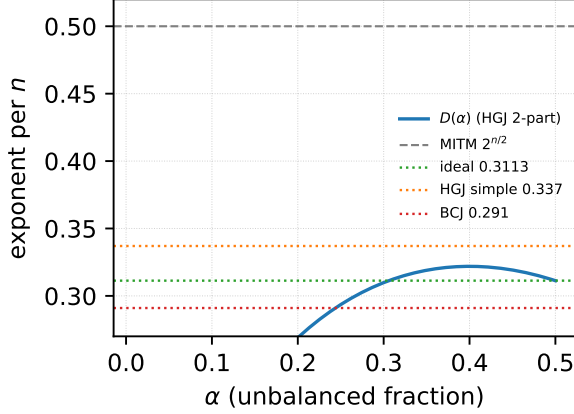


Fig. 1. The HGJ two-part exponent $D(\alpha)$ from (1) against the meet-in-the-middle $2^{n/2}$, the ideal 0.3113, the HGJ simple 0.337, and the BCJ 0.291 references.

IV. REPRODUCING THE BCJ EXPONENTS

BCJ [4] use three levels of $\{-1, 0, 1\}$ decomposition with parameters α, β, γ . The coefficient counts are

$$N_\nu(1) = \frac{1}{4} + \alpha, \quad N_\nu(-1) = \alpha, \quad (3)$$

$$N_\kappa(1) = \frac{1}{8} + \frac{\alpha}{2} + \beta, \quad N_\kappa(-1) = \frac{\alpha}{2} + \beta, \quad (4)$$

$$N_\omega(1) = \frac{1}{16} + \frac{\alpha}{4} + \frac{\beta}{2} + \gamma, \quad N_\omega(-1) = \frac{\alpha}{4} + \frac{\beta}{2} + \gamma. \quad (5)$$

With the moduli fixed by the “single decomposition” conditions, the list sizes are $L_\chi = \text{mult}(N_\chi) / \prod M$, and with $2^{h_\omega n/2}$ the cost of the bottom-level base lists (h_ω the entropy of the level- ω coefficient distribution) the running time is

$$T = \max(2^{h_\omega n/2}, L_\omega, \frac{L_\omega^2}{M_\kappa}, L_\kappa, \frac{L_\kappa^2}{M_\nu}, L_\nu, L_\nu^2 \frac{M_\nu M_\kappa M_\omega}{2^n}). \quad (6)$$

Minimising (6) numerically gives $T = 2^{0.2911n}$ at $(\alpha, \beta, \gamma) = (0.027, 0.017, 0.003)$ on a 0.0005 grid, matching BCJ’s published optimum $(0.0267, 0.0168, 0.0029)$ and $2^{0.291n}$; the base-list term is $2^{0.266n}$ there and does not bind. At $\alpha = \beta = \gamma = 0$ it recovers the May–Meurer-corrected HGJ value $2^{0.337n}$.

V. STRUCTURAL PROBES

We ran a weak (balanced) variant of the HGJ search and probed the structural hypotheses that have been proposed for the barrier.

TABLE II

COEFFICIENT-SHIFTING SEARCH. “NONTRIVIAL” COUNTS FACTORISATIONS WITH A COEFFICIENT REPRESENTED MORE THAN ONCE.

C	nontrivial factorisations	max representations
$\{0, 1\}$ (Subset Sum)	0	1
$\{\pm 1\}$ (Partition)	0	1
$\{\pm 2\}$	0	1
$\{-1, 0, 1\}$ (Equal Subset Sum)	6	2
$\{\pm 3\}$	4	2
$\{-2, \dots, 2\}$	39	3

a) *Modular coverage.*: For a planted solution, the residues $a \cdot y \bmod M$ of its $C(n/2, n/4)$ representations, with M near the representation count, are *spread* at the random-birthday level (coverage ≈ 0.6) for dissociated, density-1, and near-geometric families; only deliberately concentrated families (arithmetic, equal weights) are low, and they are easy. Dissociativity is therefore *not* the barrier.

b) *Pseudo-solutions.*: Resolving target representations by cardinality, dissociated, geometric, and near-geometric instances have a unique weight- $n/2$ solution and *zero* smaller-cardinality representations (i.e. no pseudo-solution load); pseudo-solutions appear only in the collision-heavy easy families. Pseudo-solution blow-up is therefore also *not* the barrier.

VI. THE OPEN CASES

Randolph–Wgrzycki’s worst-case tool is *coefficient shifting*: a sumset factorisation $C = C_1 + C_2$ in which coefficients acquire multiple representations, e.g. $\{0, 1\} + \{-3, -2, 1, 2\} = [\pm 3]$, where ± 2 has two representations. Searching factorisations $C_1, C_2 \subseteq [-3, 3]$ (Table II), the open cases $\{0, 1\}$, $\{\pm 1\}$ (Partition), and $\{\pm 2\}$ are, within this range, *exactly* those with no nontrivial factorisation: every coefficient keeps a single representation. Sets such as $\{-1, 0, 1\}$ and $\{\pm 3\}$ admit a nontrivial factorisation and obtain the surplus that shifting needs.

This reproduces the frontier within the searched range: the obstacle for $\{0, 1\}$ is the absence of a nontrivial sumset factorisation (note that $\{0, 1\}$ does contain 0, so the presence of 0 alone does not help), so the worst-case representation technique has no surplus to exploit. A worst-case $2^{n/3}$ algorithm for $\{0, 1\}$ must therefore use a *different* mechanism.

VII. CONCLUSION

The representation technique already reaches $2^{0.291n}$ for random hard knapsacks. Its limits are not a sub-solver or a class of hard-looking instances (those are representation-friendly), but the average-case nature of its modular guarantees and the lack of coefficient shifting for $\{0, 1\}$, $\{\pm 1\}$, $\{\pm 2\}$. A worst-case $2^{n/3}$ bound for exact subset sum requires a new mechanism for these sets, or a conditional lower bound.

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